Diagonalization

Geometric Algorithms

CAS CS 132

Objectives

» Finish our discussion on the characteristic polynomial

» Motivate diagonalization via linear dynamical systems and changes of coordinate systems

» Describe how to diagonalize a matrix

Keywords

multiplicity

similar matrices

diagonalizable matrices

change of basis

eigenbasis

Recap: Characteristic Polynomial

Recall: Determinants and Invertibility

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$\det(A - \lambda I) = 0 \quad \equiv \quad (A - \lambda I)\mathbf{x} = \mathbf{0}$ **has nontrivial solutions**

$\equiv \quad \lambda$ is an eigenvalue of A

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polynomial in λ

$\equiv$ λ is an eigenvalue of A

How To: Finding Eigenvalues

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$$\det(A - \lambda I)$$

Example

$$A = \begin{bmatrix} 1 & -1 \\ 7 & -3 \end{bmatrix}$$

An Observation: Multiplicity

$$\lambda^{1}(\lambda-1)^{2}(\lambda-4)^{1}$$

multiplicities

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In the examples so far, we've seen a number appear as a root multiple times

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This is called the **algebraic multiplicity** of the root

Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

Thm. The dimension of the eigenspace of A for the eigenvalue λ is *at most* the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Practice Problem

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Determine the eigenvalues and an eigenbasis for the above matrix

Answer

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Motivating Diagonalization via Linear Dynamical Systems

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When can we describe any vector in $\mathbb{R}^n$ as a unique linear combination of eigenvectors of A?

Recall: Linear Dynamical Systems

$$\begin{aligned} \mathbf{v}_1 &= A\mathbf{v}_0 \\ \mathbf{v}_2 &= A\mathbf{v}_1 = A^2\mathbf{v}_0 \\ \mathbf{v}_3 &= A\mathbf{v}_2 = A^3\mathbf{v}_0 \\ \mathbf{v}_4 &= A\mathbf{v}_3 = A^4\mathbf{v}_0 \\ &\vdots \end{aligned}$$

A **linear dynamical system** describes a sequence of **state vectors** starting at $\mathbf{v}_0$

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$$\vdots$$

multiplying by A changes the state

A **linear dynamical system** describes a sequence of **state vectors** starting at $\mathbf{v}_0$

demo

Eigenbases and Closed-Form solutions

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Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

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then

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$$A^k\mathbf{v}_0 = \underbrace{\alpha_1 \lambda_1^k \mathbf{b}_1 + \alpha_2 \lambda_2^k \mathbf{b}_2 + \alpha_3 \lambda_3^k \mathbf{b}_3}_{\text{closed-form solution}}$$

Application: Eigenbases and Limiting Behavior

Thm. If A has an eigenbasis with eigenvalues

$$\lambda_1 \geq \lambda_2 \ldots \geq \lambda_k$$

then $\mathbf{v}_k \sim \lambda_1^k \mathbf{u}$ for some vector $\mathbf{u}$

The system grows exponentially in λ_1

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Given a basis $\mathscr{B}$ for $\mathbb{R}^n$, we only need to know how $A \in \mathbb{R}^n$ behaves on $\mathscr{B}$

Sometimes, A behaves simply on $\mathscr{B}$, as in the case of *eigenbases*

What we're really doing is changing our coordinate system to expose a behavior of A

Recap: Change of Basis

Recall: Bases define Coordinate Systems

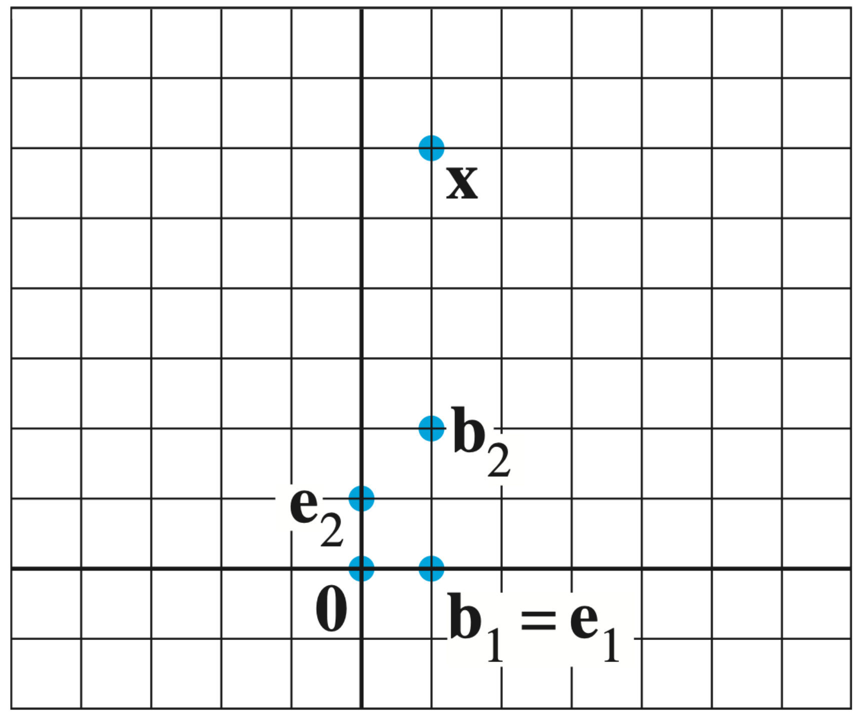


FIGURE 1 Standard graph paper.

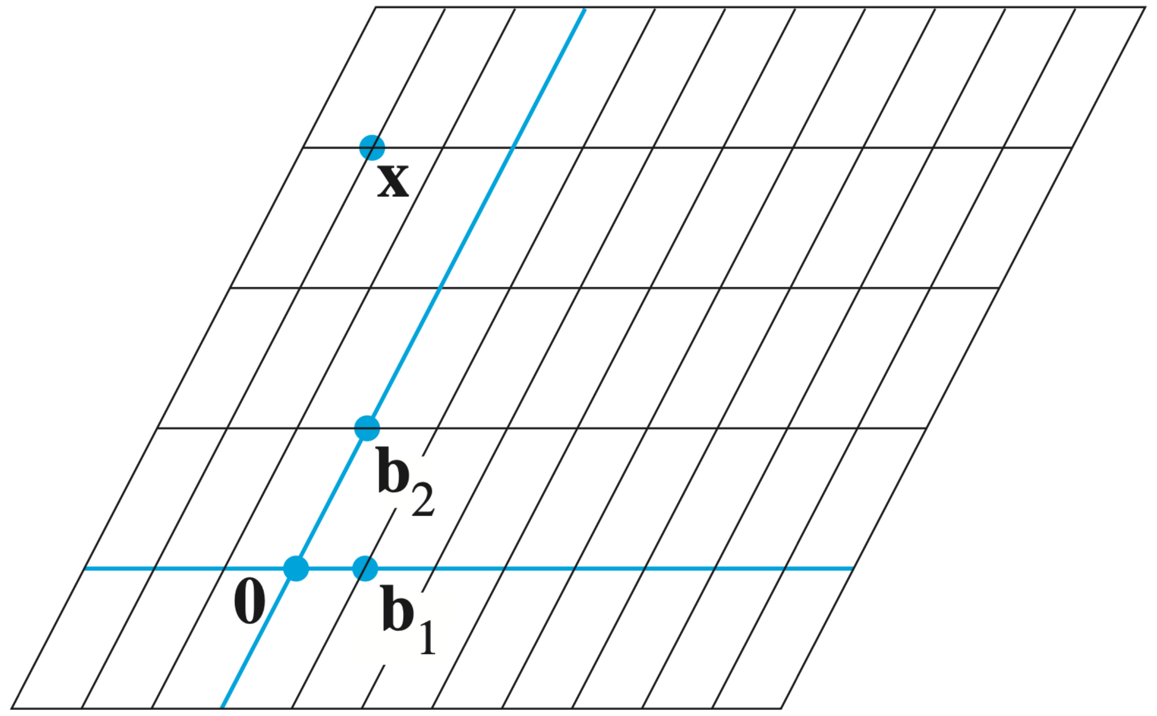
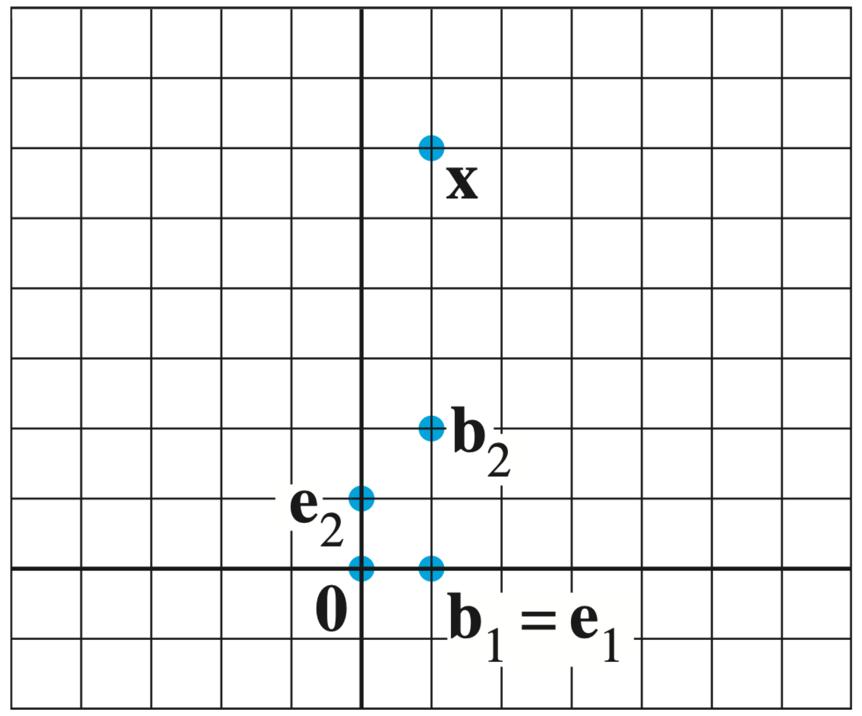


FIGURE 2 $\mathcal{B}$-graph paper.

Linear Algebra and its Applications (Lay, Lay, McDonald)

Recall: Bases define Coordinate Systems



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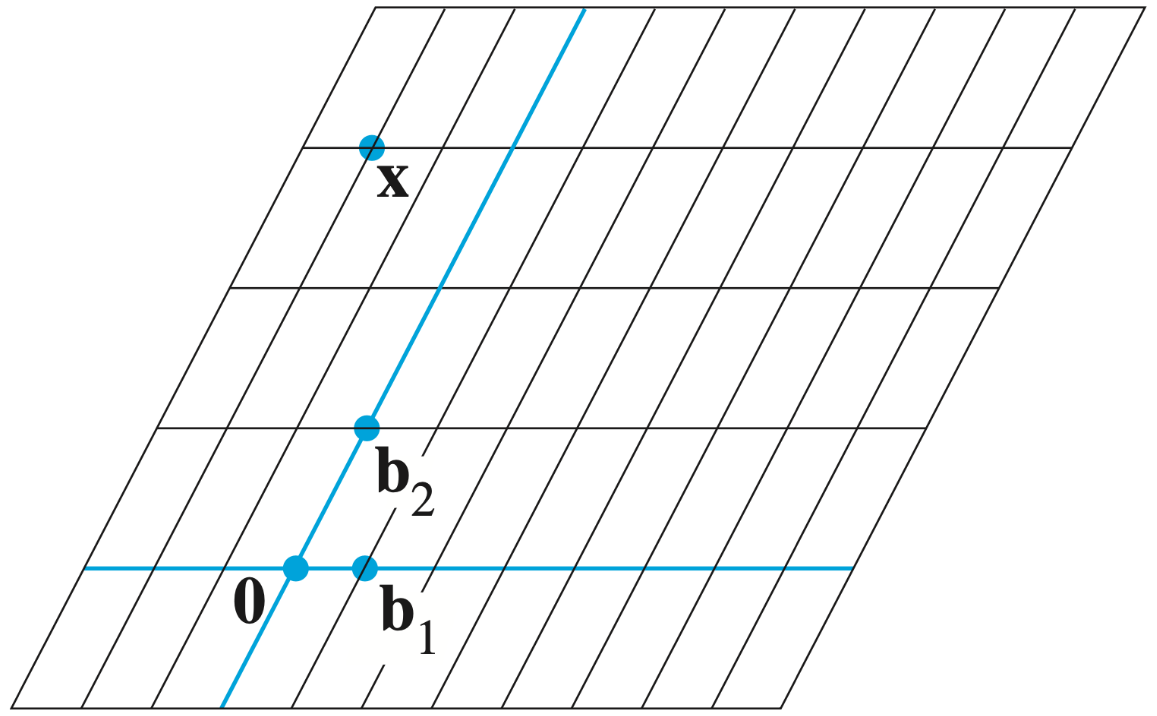

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Given a basis $\mathscr{B}$ of $\mathbb{R}^n$, there is **exactly one way** to write every vector as a linear combination of vectors in $\mathscr{B}$

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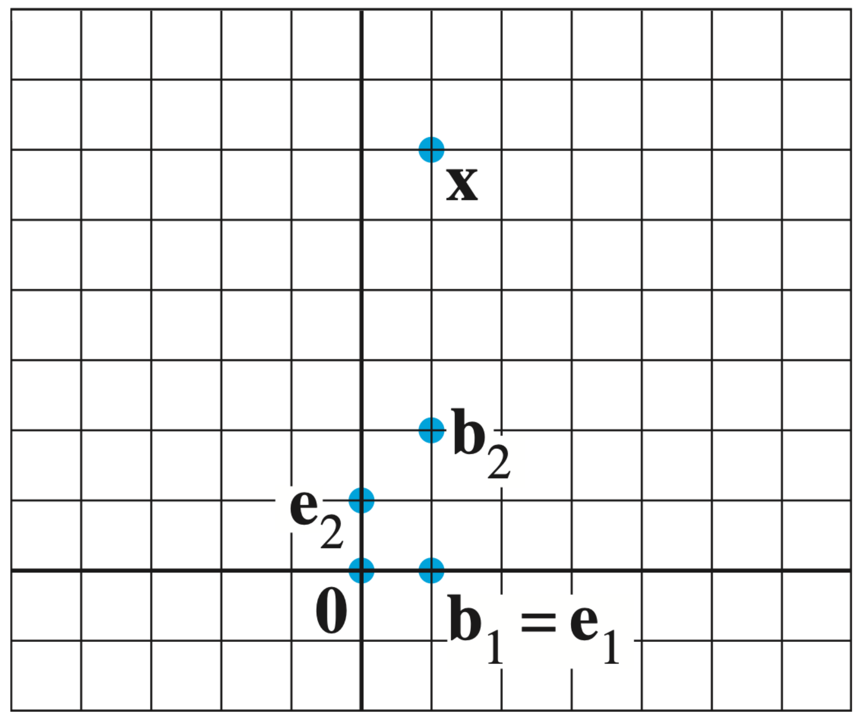

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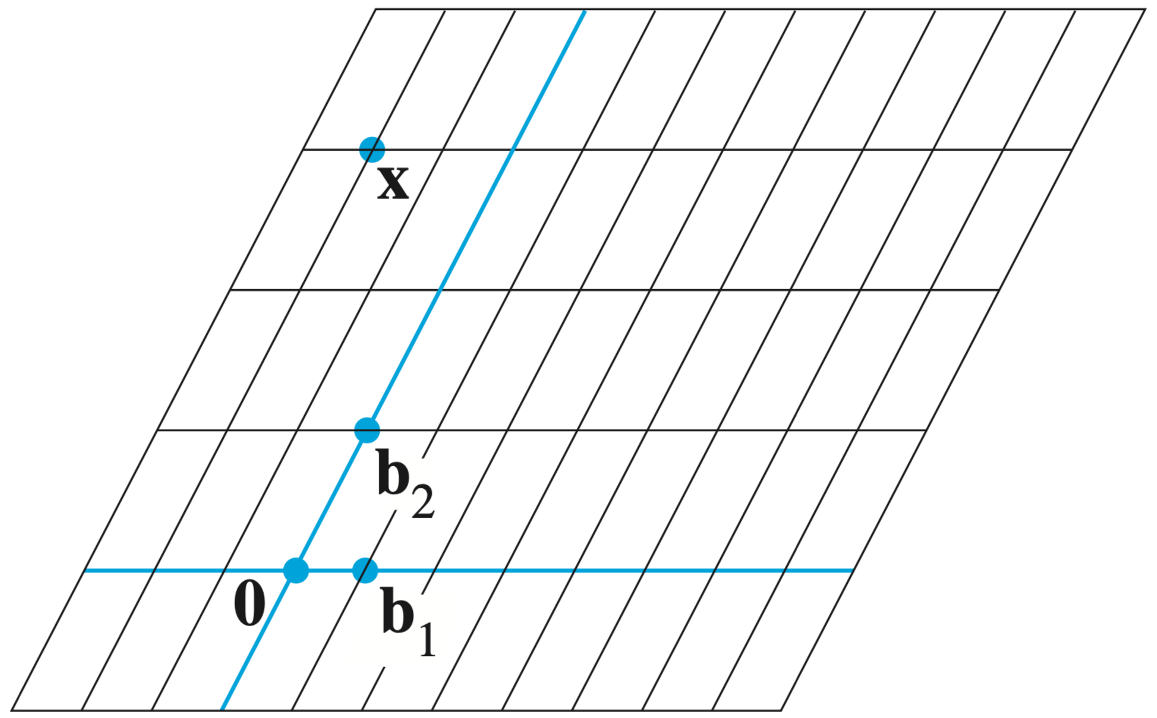

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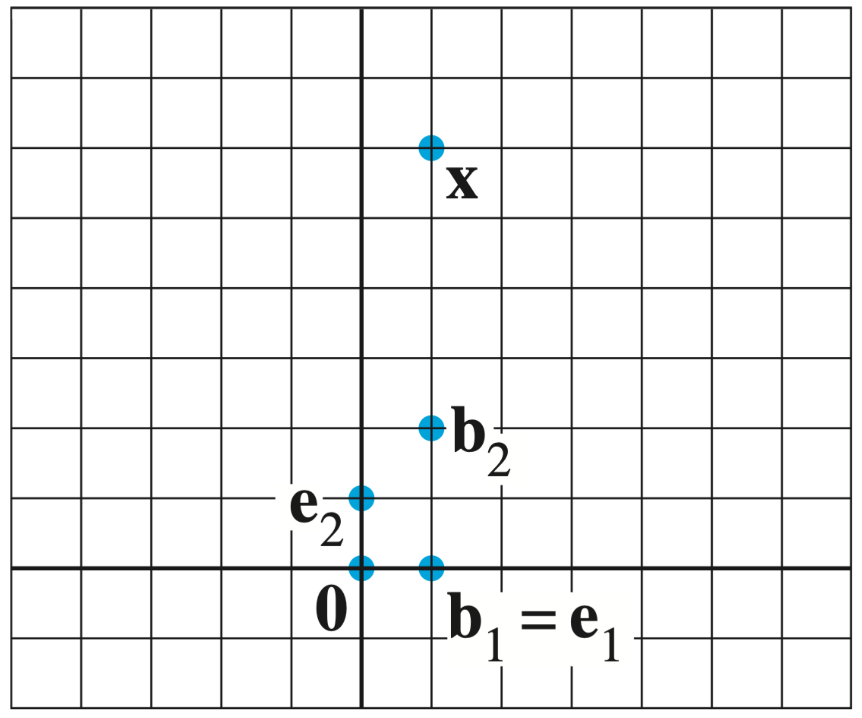

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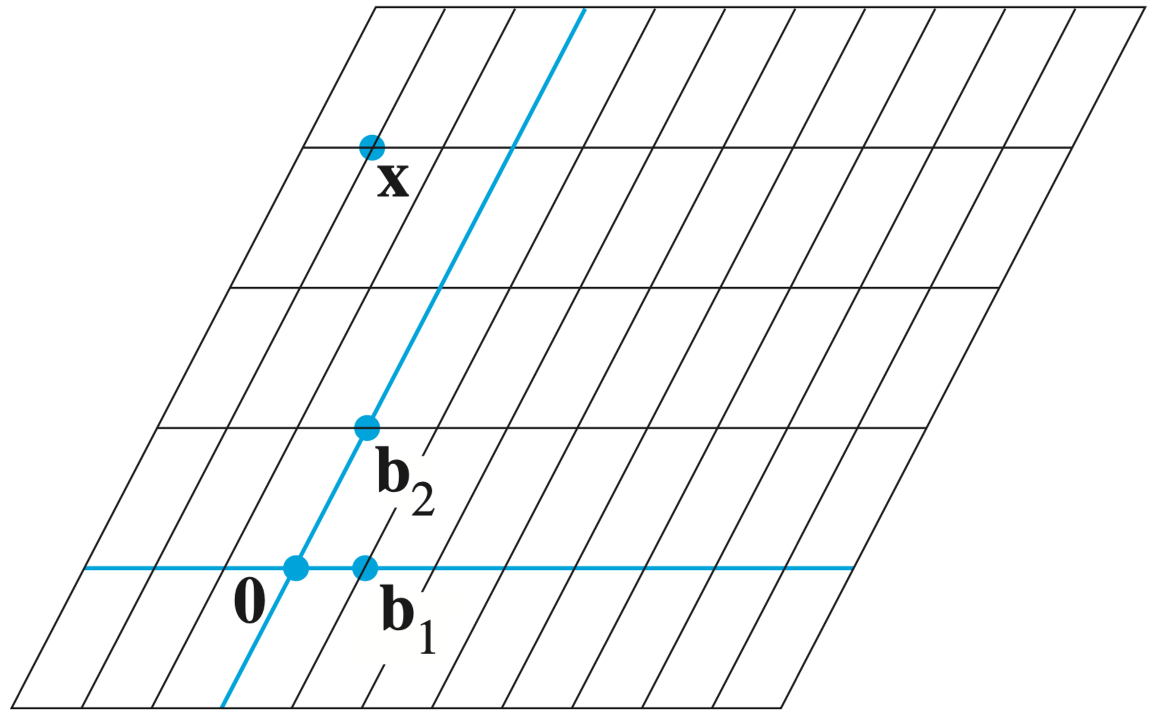

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Every basis provides a way to write down *coordinates* of a vector

$\mathscr{B}$ defines a "different grid for our graph paper"

Recall: Coordinate Vectors

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Let $\mathbf{v}$ be a vector in a $\mathbb{R}^n$ and let $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ be a basis of $\mathbb{R}^n$ where $\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n$

Df. The **coordinate vector of $\mathbf{v}$ relative to** $\mathscr{B}$ is

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Df. The **coordinate vector of $\mathbf{v}$ relative to $\mathscr{B}$** is

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Question

We know that if a $n \times n$ matrix $B = [\mathbf{b}_1\ \mathbf{b}_2\ \dots\ \ \mathbf{b}_n] \in \mathbb{R}^{n \times n}$ is invertible, then the columns of B form a basis $\mathscr{B}$ of $\mathbb{R}^n$

What is the matrix that implements the transformation

$$\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

where $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n$*?*

Change of Basis Matrix

Thm. If $\mathscr{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$ form a basis of $\mathbb{R}^n$, then

$$[\mathbf{x}]_{\mathscr{B}} = [\mathbf{b}_1\ \mathbf{b}_2\ \ \ldots\ \ \mathbf{b}_n]^{-1}\mathbf{x}$$

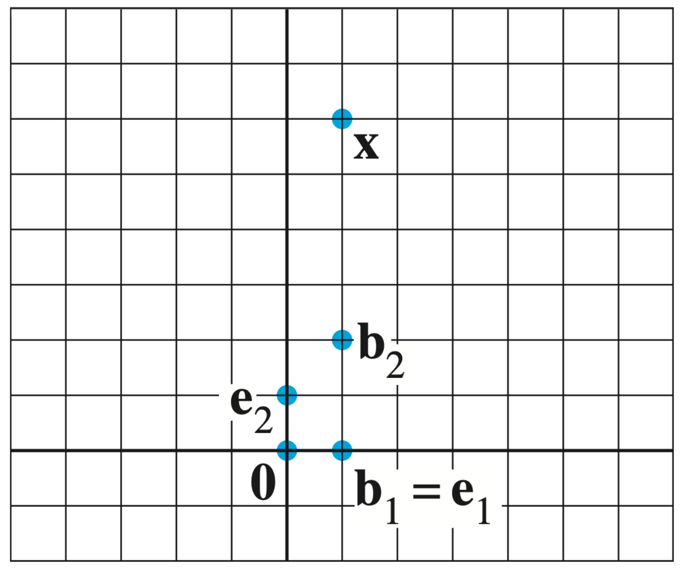
Matrix inverses perform changes of bases

How To: Change of Basis

Q. Given a basis $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ of $\mathbb{R}^n$, find the matrix which implements $\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}}$

A. Construct the matrix $[\mathbf{b}_1 \ \mathbf{b}_2 \ \dots \ \mathbf{b}_n]^{-1}$

Example



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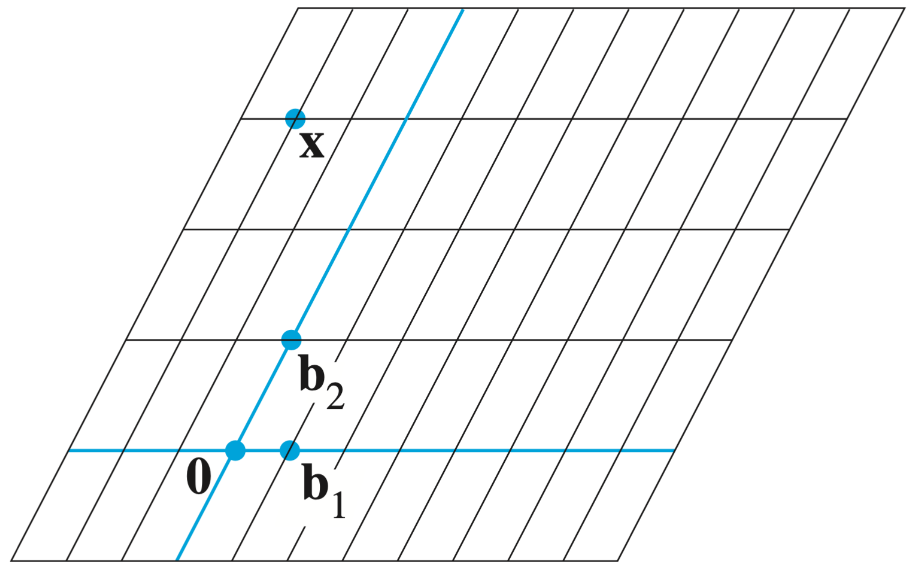

FIGURE 2 $\mathcal{B}$-graph paper.

Write the change-of-bases matrix for the basis $\left(\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}\right)$

Diagonalization

Diagonal Matrices

ex. $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Def. A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonal** if

$i \neq j$ if and only if $A_{ij} = 0$

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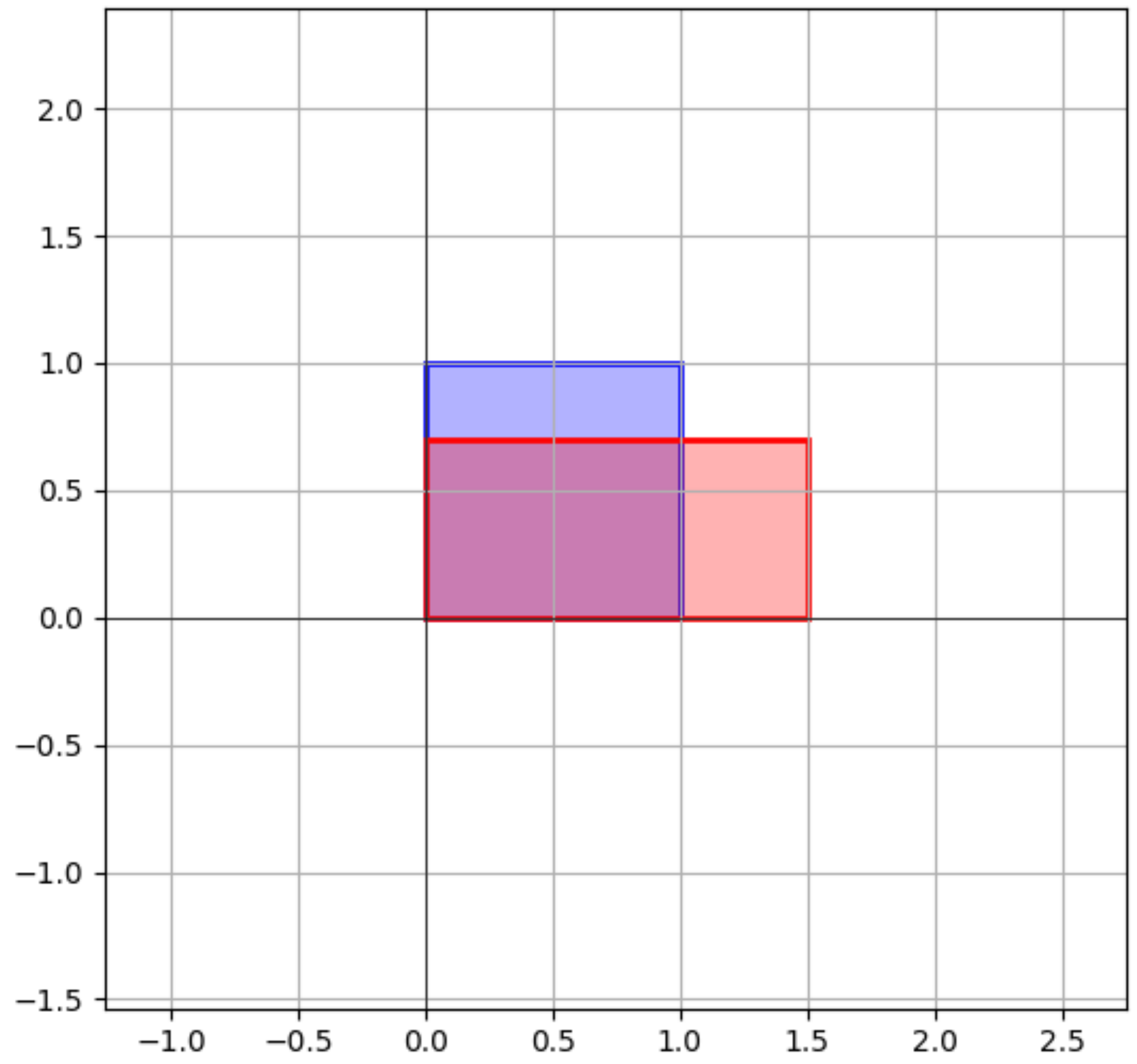
Diagonal matrices are scaling matrices

Recall: Unequal Scaling

The scaling matrix *affects each component of a vector in a simple way*

The diagonal entries <u>scale</u> each corresponding entry

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.5x \\ 0.7y \end{bmatrix}$$



$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

When do matrices "behave" like scaling matrices "up to" change of basis?

Scaling and Eigenvectors

Scaling and Eigenvectors

The idea. Matrices behave like scaling matrices on eigenvectors

Can we expose this behavior in terms of a matrix factorization?

Recall: Matrix Factorization

A **factorization** of a matrix A is an equation which expresses A as a product of one or more matrices, e.g.,

$$A = PBP^{-1}$$

Similar Matrices

$$A = PBP^{-1}$$

Def. A matrix A is **similar** to a matrix B if there is some invertible matrix P such that $A = PBP^{-1}$

A and B are the same up to a change of basis

Similar Matrices and Eigenvalues

Thm. Similar matrices have the *same eigenvalues*

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Def. A matrix A is **diagonalizable** if it is similar to a diagonal matrix

There is an invertible matrix P *and diagonal matrix* D *such that* $A = PDP^{-1}$

Diagonalizable matrices are the same as scaling matrices up to a change of basis

Important: Not all Matrices are Diagonalizable

This is very different from the LU factorization

We will need to figure out which matrices are diagonalizable

Application: Matrix Powers

only take the power of B

Thm. If $A = PBP^{-1}$, then $A^k = PB^kP^{-1}$

It may be easier to take the power of B (as in the case of diagonal matrices)

But how do we find the diagonalization...

Diagonalization and Eigenvectors

Suppose we have a diagonalization

$$A = PDP^{-1}$$

What do we know about it?

Columns of P are eigenvectors

$$A = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3]^{-1}$$

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And the entries of D are the **eigenvalues** associated to each eigenvector

Columns of P are eigenvectors

$$A = \overset{\text{eigenbasis}}{\boxed{[\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]}} \overset{\text{eigenvalues}}{\begin{bmatrix} \boxed{\lambda_1} & 0 & 0 \\ 0 & \boxed{\lambda_2} & 0 \\ 0 & 0 & \boxed{\lambda_3} \end{bmatrix}} [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

In fact, the columns of P form an **eigenbasis** of $\mathbb{R}^n$ for A

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A diagonalization exposes a lot of information about A

The Diagonalization Theorem

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Thm. A matrix is diagonalizable if and only if it has an eigenbasis

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Thm. A matrix is diagonalizable if and only if it has an eigenbasis

We can use the same recipe to go in the other direction, given an eigenbasis, we can **build a diagonalization**

Diagonalizing a Matrix

High Level

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The matrix P^{-1} is a change of basis to this eigenbasis of A

Step 1: Eigenvalues

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

Find all the eigenvalues of A

Find the roots of $\det(A - \lambda I)$

e.g. $\det(A - \lambda I) = -(\lambda - 1)(\lambda + 2)^2$

Step 2: Eigenvectors

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

`Find **bases** of the corresponding eigenspaces`

e.g.

$$\mathrm{Nul}(A - I) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A + 2I) = \mathrm{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Step 3: Construct P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$\text{Nul}(A - I) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\text{Nul}(A + 2I) = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

If there are n eigenvectors from the previous step they form an **eigenbasis**

Build the matrix with these vectors as the columns

e.g.

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 5: Construct D

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Build the matrix with eigenvalues as diagonal entries

Note the order. It should be the same as the order of columns of P

e.g.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Step 6: Invert P

Find the inverse of P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

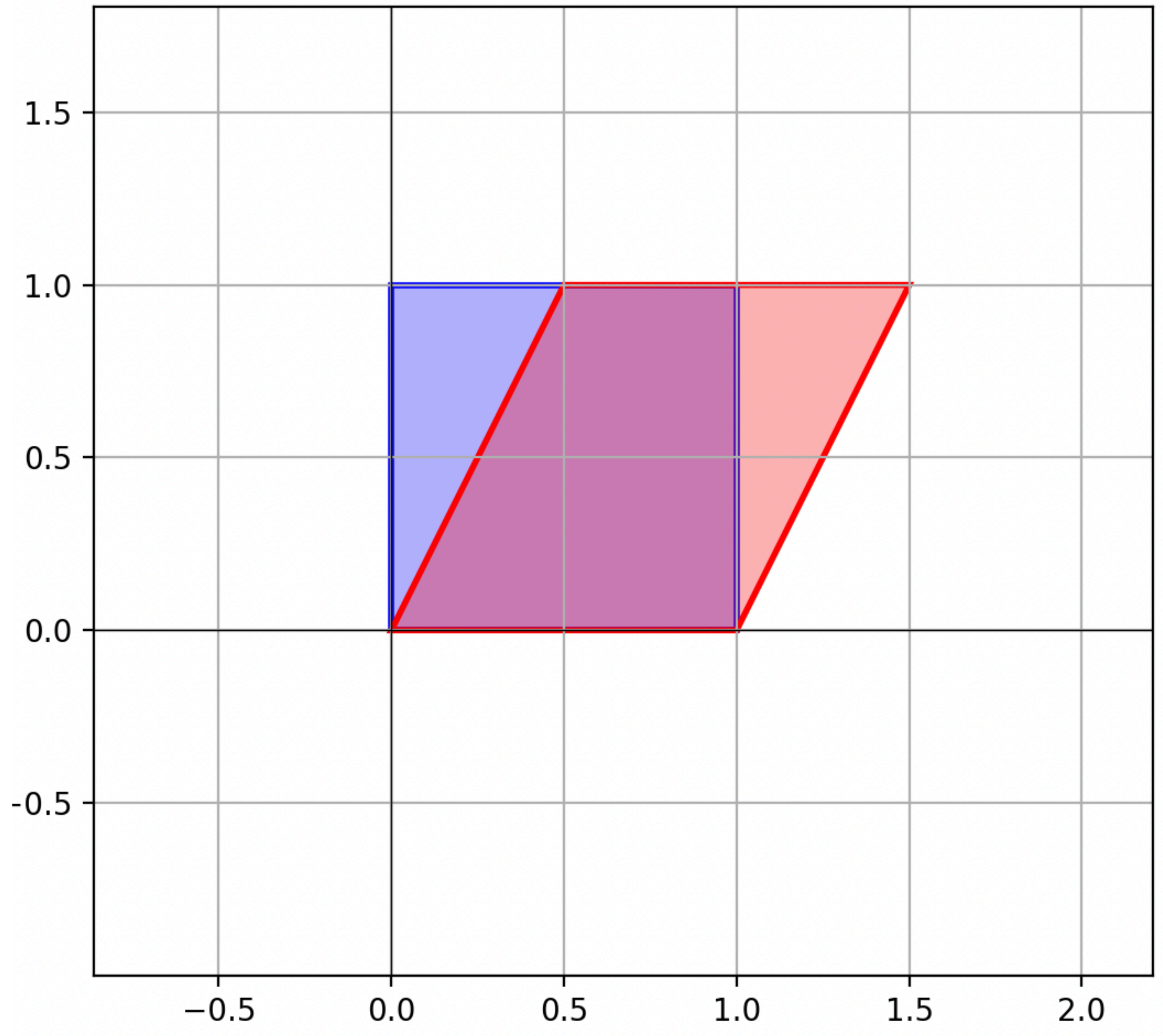
Putting it Together

$$\underset{A}{\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}} = \underset{P}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}} \underset{D}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}} \underset{P^{-1}}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}}$$

We know how to do every step, its a matter of putting it all together

Example of Failure: Shearing $A = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$

The shearing matrix has a single eigenvalue with an eigenspace of dimension 1



We can't build an eigenbasis of $\mathbb{R}^2$ for A

In other words, A is not diagonalizable

Important case: Distinct Eigenvalues

$$\begin{bmatrix} 1 & -3 & 4 & 2 \\ 0 & -2 & 3 & -1 \\ 0 & 0 & 10 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Thm. If a matrix in $\mathbb{R}^{n\times n}$ has n distinct eigenvalues, then it is diagonalizable

This is because eigenvectors with distinct eigenvalues are *linearly independent*

Example

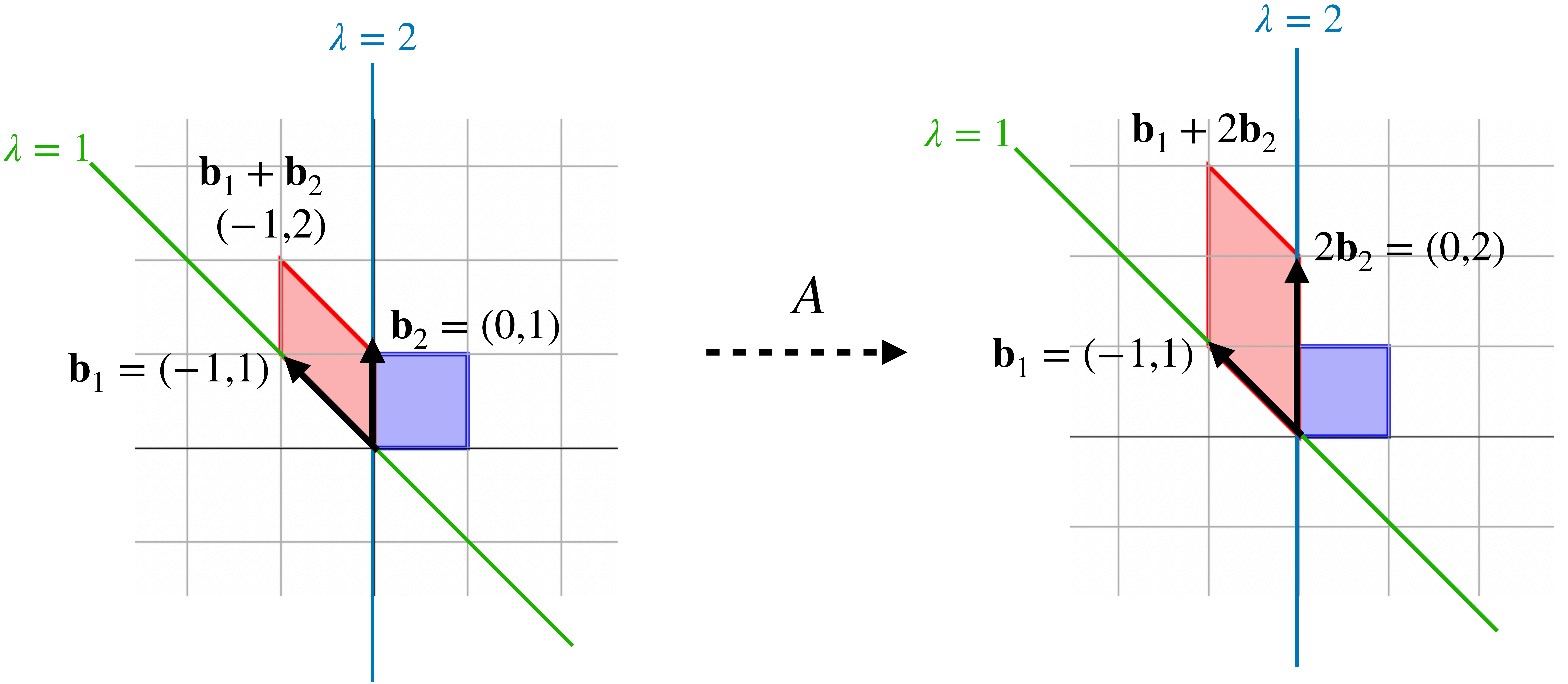
$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$$

Find a diagonalization of the above matrix

The Picture

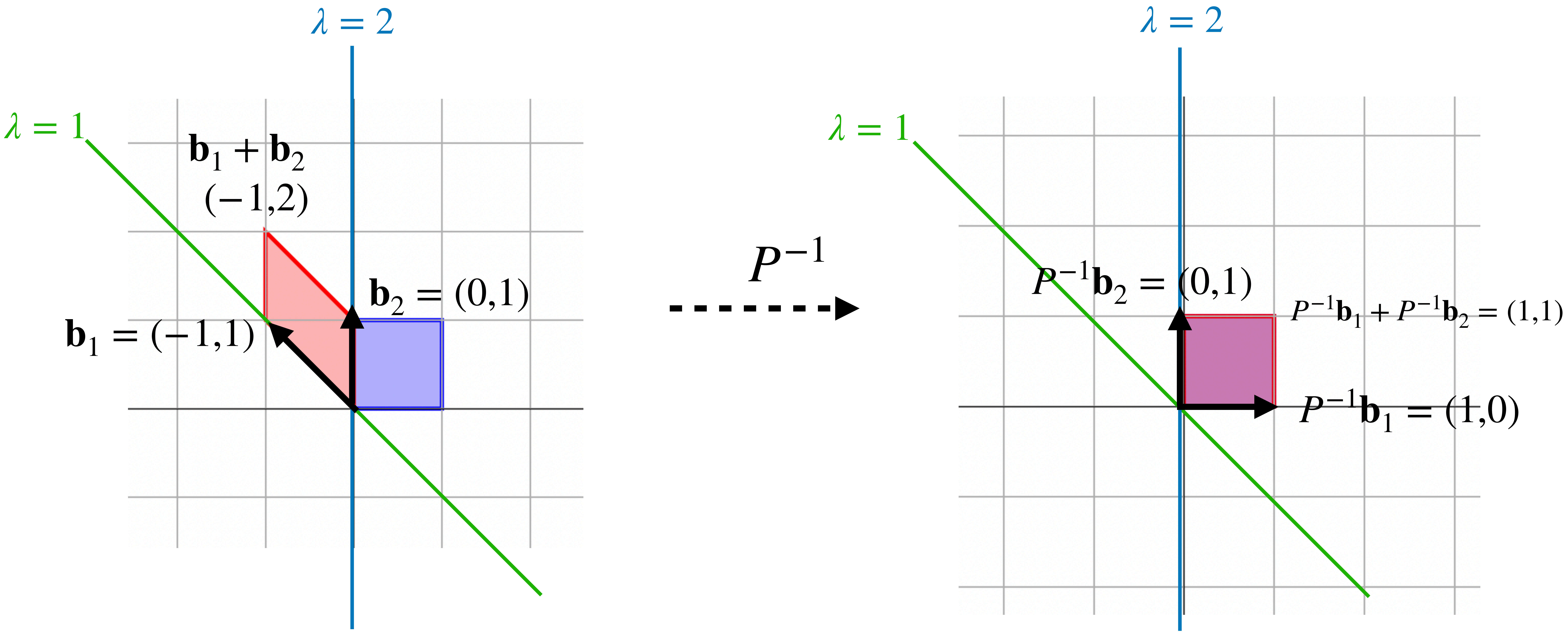
Example (Geometric)

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$$



Example (Geometric)

$$P^{-1} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$

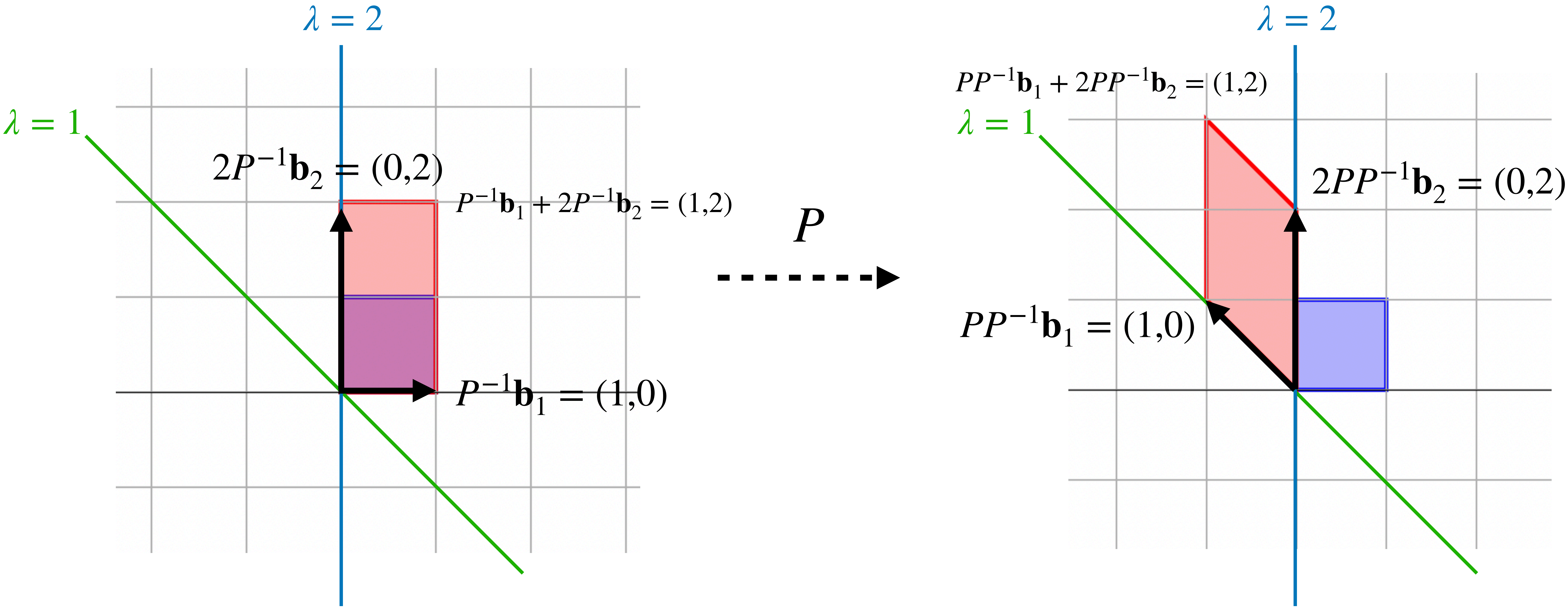


Example (Geometric)

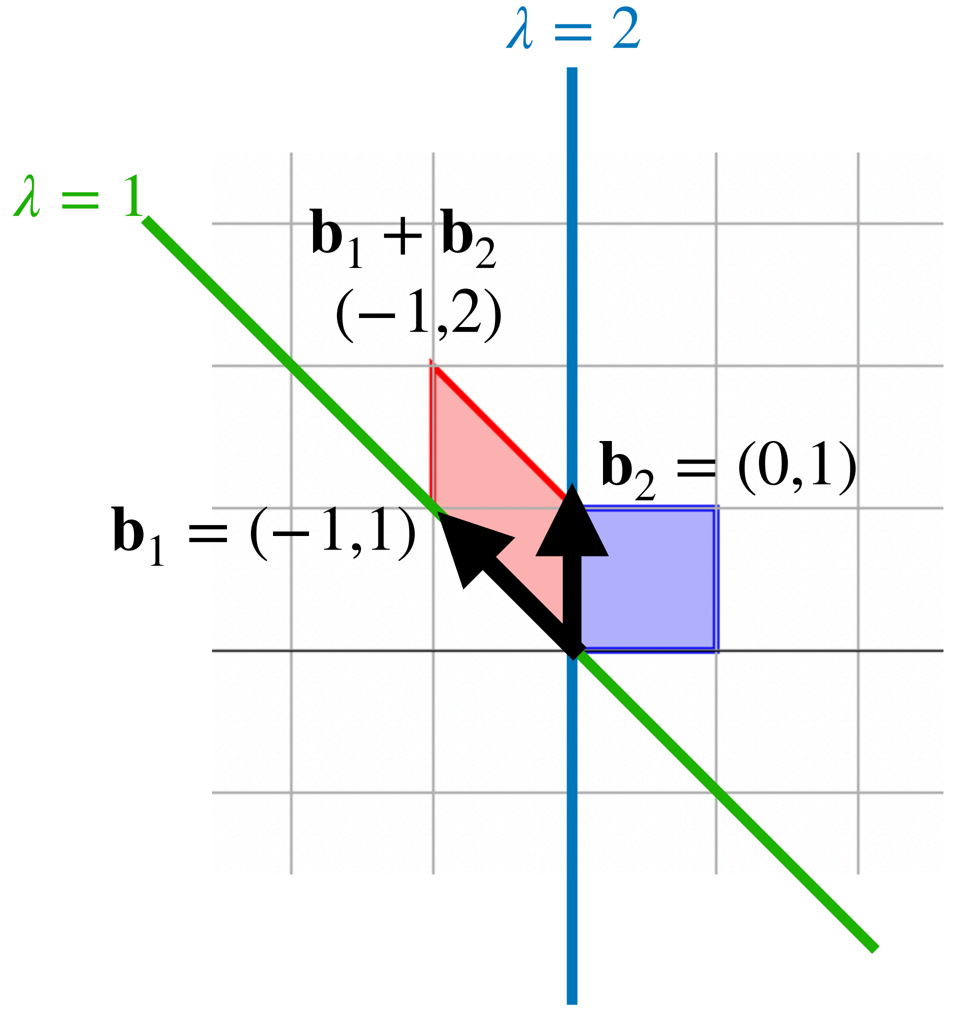
$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Example (Geometric)

$$P = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$$



Example (Geometric)



$$A = PDP^{-1}$$

$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$